

# Physical Computing

## Syllabus & Overview

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# Description

– Module phycom

\* <https://www.fhnw.ch/de/studium/module/9880131>



## # Language

- Inputs in class are in German (DE)
- Course materials are in English (EN)
- Your contribution can be in DE or EN



## # Credits

- 3 ECTS, 2 \* 5 days, 8 + 1 hours a day
- 2 joker hours, ask in advance
- Pass/fail, no grades



## # Date

- Week 1, KW24, 08.06. – 12.06.2026
- Week 2, KW25, 15.06. – 19.06.2026



## # Time

- 09:15 - 12:00, 3 hours in the morning
- 13:15 - 18:00, 5 hours in the afternoon
- 15 minute breaks, roughly once an hour



## # Location

- Input, usually at the classroom, Room 4.513
- Making, usually at the makerspace, Room 4.125





## # Presence

- > Students are expected to be present for the entire two weeks
- The course consists of inputs and hands-on making sessions
- We will use specific tools available in a typical makerspace
- We count on you all to share your experience/perspective



## # Program Week 1

| Day | Input                     | Making        |
|-----|---------------------------|---------------|
| Mon | Intro & DIY-Electronics   | Soldering     |
| Tue | Embedded Programming      | Tinkering     |
| Wed | Design & Ideation         | Prototyping   |
| Thu | Digital Fabrication in 2D | Laser-cutting |
| Fri | Digital Fabrication in 3D | 3D-printing   |



## # Program Week 2

| Day | Input                   | Making     |
|-----|-------------------------|------------|
| Mon | PCB-Layout              | Etching    |
| Tue | Documentation & Sharing | Exploring  |
| Wed | DIY-Project Kick-off    | Working    |
| Thu | DIY-Project Inputs      | Working    |
| Fri | DIY-Project Inputs      | Presenting |



## # Program Specials

> At HTU, all participants are welcome

- Week 1, Mon, 07:45, welcome coffee
- Week 1, Fri, 12:00, free open lunch
- Week 2, Fri, 16 – 18:00, grill party



## # Topics

0. Syllabus & Overview
1. DIY-Electronics & Soldering
2. Embedded Programming
3. Design, Ideation & Prototyping
4. Digital Fabrication in 2D
5. Digital Fabrication in 3D
6. PCB-Layout & Production
7. Documentation & Sharing
8. DIY-Project & Presentation



## # Structure

Each topic follows this structure

- Motivation, why this is relevant, by example
- Basics, what you have to know, to get started
- Advanced, what brings you further, after basics
- Materials, what we provide to do hands-on tasks
- Tools, which tools you can use to get experience
- Skills, what you can learn by doing it yourself
- Resources, where you find more, quality info



## # Materials

- Template repo, as a starting point
- Box of tools per team, to be returned
- Making materials, to be used responsibly

\* <https://github.com/tamberg/fhnw-phycom>



## # Audience

- This course can be attended by students from both, HSI and HTU
- Basic knowledge of either programming or electronics might help
- If you are not sure if this course is for you, consider contacting us





## # Contact

- Contact us via MS Teams or email  
[thomas.amberg@fhnw.ch](mailto:thomas.amberg@fhnw.ch)

